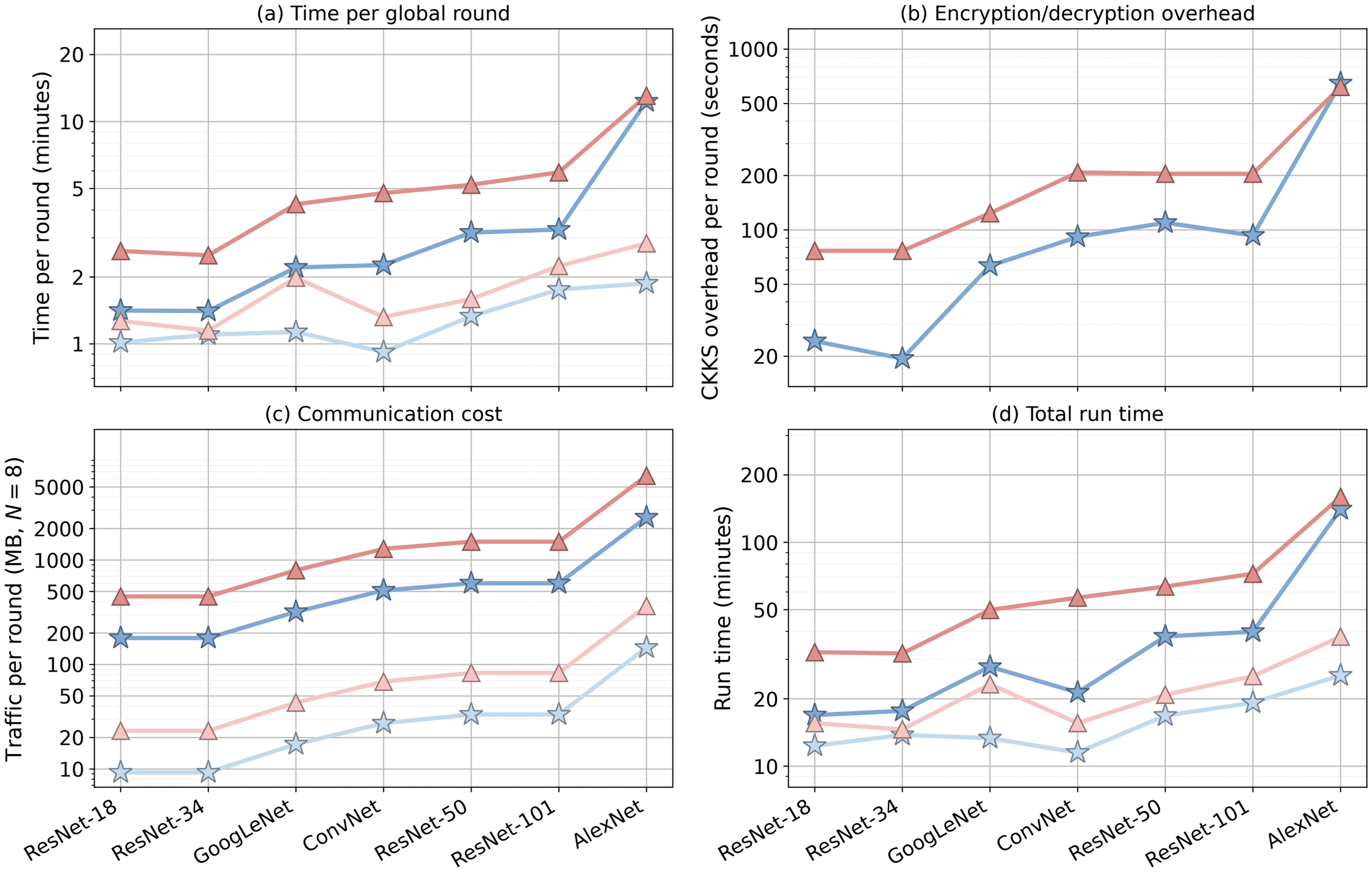
(a) Time per global round
Time per round (minutes)
20
10
5
2
1
(b) Encryption/decryption overhead
CKKS overhead per round (seconds)
1000
500
200
100
50
20
(c) Communication cost
Traffic per round (MB, $N = 8$)
5000
2000
1000
500
200
100
50
20
10
(d) Total run time
Run time (minutes)
200
100
50
20
10
ResNet-18
ResNet-34
GoogLeNet
ConvNet
ResNet-50
ResNet-101
AlexNet